21 Oct 2026 00:09:26 - if97_steam_tables.sm
Created using a free version of SMath Studio

IAPWS-IF97 High-Precision Steam Tables & Thermodynamic Properties

2026-10-19

IAPWS-IF97 Water and Steam Properties Plugin for SMath Studio

Industrial Formulation 1997 for General & Scientific Power-Plant Calculations

Native Assembly: IF97Wrapper.dll (Plugin GUID: bdb8d1d9-11bf-4f8b-aca4-c1fa3f6e13ad, v1.1.1.0)

Strict SI Unit Guarantee: Pressures in [Pa], Temperatures in [K], Energies in [J/kg] (m^2/s^2), Entropies in [J/(kg*K)]

The International Association for the Properties of Water and Steam (IAPWS) IF97 formulation

is the international industrial standard for calculating thermodynamic and transport properties

of water and steam across power plant engineering, turbomachinery design, and thermal processes.

Key Architectural Highlights of the IF97 SMath Plugin (v1.1.1.0 .NET 2.0 CLR):

1. Pure SI Units: Eliminates unit conversion pitfalls. All inputs accept standard SMath SI units

   or bare numbers (implicitly SI); all outputs return strictly in standard SMath SI units.

2. Direct Fast Forward Functions: Sub-microsecond evaluation of h, s, u, rho, v, cp, cv, w, mu, tc, Pr.

   Standard argument order is (T, p) with temperature in [K] and pressure in [Pa].

3. Direct Backward Functions: Analytical backward equations T(p,h), rho(p,h), T(p,s), rho(p,s), p(h,s), T(h,s).

4. Wet Steam Two-Phase Region: Saturated liquid/vapor lines, latent heat hfg, and two-phase mixtures.

5. Standardized 46-Function API: Clean IF97_<property>_<inputs> nomenclature with strict base SI unit emission.

Section 1: Atmospheric & Saturation Boundary Calculations

Verification of saturation temperature, saturation pressure, surface tension, and saturated liquid/vapor lines.

Standard atmospheric pressure: P_atm = 101325 Pa (1.01325 bar, 1 atm)

$P_atm := 101325\ \mathrm{Pa}$

Atmospheric boiling point: T_boil_1atm = 373.1243 K (99.974 °C)

$T_boil_1atm := \mathrm{IF97_Tsat_p}(P_atm) = 373.1243\ \mathrm{K}$

Boiling temperature benchmark: T_100C = 373.15 K (100.0 °C)

$T_100C := 100 + 273.15\ \mathrm{K}$

Saturation vapor pressure at 100 °C: P_sat_100C = 101417.98 Pa (1.01418 bar)

$P_sat_100C := \mathrm{IF97_psat_T}(T_100C) = 1.0142 \cdot 10^{5}\ \mathrm{Pa}$

Water-steam surface tension at 100 °C: sigma_100C = 0.05891 N/m

$sigma_100C := \mathrm{IF97_sigma_T}(T_100C) = 0.0589\ \mathrm{m\ Pa}$

Not for commercial use

Created using a free version of SMath Studio

Reference saturation pressure: P_sat_1MPa = 1.0 MPa (10.0 bar, 1e6 Pa)

$P_sat_1MPa := 1000000\ Pa$

Saturation temperature at 1 MPa: T_sat_1MPa = 453.0356 K (179.886 °C)

$T_sat_1MPa := IF97_Tsat_p(P_sat_1MPa) = 453.0356\ K$

Saturated liquid enthalpy at 1 MPa: h_f_1MPa = 762682.8 J/kg (762.68 kJ/kg)

$h_f_1MPa := IF97_hliq_p(P_sat_1MPa) = 7.6268 \cdot 10^{5}\ Gy$

Saturated vapor enthalpy at 1 MPa: h_g_1MPa = 2777119.5 J/kg (2777.12 kJ/kg)

$h_g_1MPa := IF97_hvap_p(P_sat_1MPa) = 2.7771 \cdot 10^{6}\ Gy$

Latent heat of vaporization at 1 MPa: h_fg_1MPa = 2014436.7 J/kg (2014.44 kJ/kg)

$h_fg_1MPa := IF97_hfg_p(P_sat_1MPa) = 2.0144 \cdot 10^{6}\ Gy$

Saturated liquid entropy at 1 MPa: s_f_1MPa = 2138.43 J/(kg·K) (2.1384 kJ/(kg·K))

$s_f_1MPa := IF97_sliq_p(P_sat_1MPa) = 2138.4314\ \frac{Gy}{K}$

Saturated vapor entropy at 1 MPa: s_g_1MPa = 6584.98 J/(kg·K) (6.5850 kJ/(kg·K))

$s_g_1MPa := IF97_svap_p(P_sat_1MPa) = 6584.979\ \frac{Gy}{K}$

Saturated liquid density at 1 MPa: rho_f_1MPa = 887.13 kg/m³

$rho_f_1MPa := IF97_rholiq_p(P_sat_1MPa) = 887.1275\ \frac{kg}{m^3}$

Saturated vapor density at 1 MPa: rho_g_1MPa = 5.145 kg/m³

$rho_g_1MPa := IF97_rhovap_p(P_sat_1MPa) = 5.1454\ \frac{kg}{m^3}$

Saturated liquid specific volume at 1 MPa: v_f_1MPa = 0.001127 m³/kg

$v_f_1MPa := \frac{1}{rho_f_1MPa} = 0.0011\ \frac{m^3}{kg}$

Saturated vapor specific volume at 1 MPa: v_g_1MPa = 0.19435 m³/kg

$v_g_1MPa := \frac{1}{rho_g_1MPa} = 0.1943\ \frac{m^3}{kg}$

Saturated liquid dynamic viscosity at 1 MPa: mu_f_1MPa = 1.5049e-4 Pa·s

$mu_f_1MPa := IF97_muliq_p(P_sat_1MPa) = 0.0002\ s\ Pa$

Saturated vapor dynamic viscosity at 1 MPa: mu_g_1MPa = 1.4981e-5 Pa·s

$mu_g_1MPa := IF97_muvap_p(P_sat_1MPa) = 1.4981 \cdot 10^{-5}\ s\ Pa$

Saturated liquid thermal conductivity at 1 MPa: tc_f_1MPa = 0.6713 W/(m·K)

$tc_f_1MPa := IF97_tcliq_p(P_sat_1MPa) = 0.6713\ \frac{N}{s\ K}$

Saturated vapor thermal conductivity at 1 MPa: tc_g_1MPa = 0.03481 W/(m·K)

$tc_g_1MPa := IF97_tcvap_p(P_sat_1MPa) = 0.0348\ \frac{N}{s\ K}$

Not for commercial use

Created using a free version of SMath Studio

Visualization 1: Water Vapor Pressure Saturation Curve Psat(T)

Clausius-Clapeyron exponential vapor pressure rise spanning 10 °C up to near-critical 370 °C (210 bar).

Temperature nodes for vapor pressure curve: [10 ... 370] °C (283.15 ... 643.15 K)

$$T_sat_nodes := \begin{bmatrix} 283.15 \\ 303.15 \\ 323.15 \\ 343.15 \\ 363.15 \\ 373.15 \\ 393.15 \\ 423.15 \\ 453.15 \\ 483.15 \\ 523.15 \\ 573.15 \\ 623.15 \\ 643.15 \end{bmatrix} \text{K}$$

Saturation vapor pressures: [1228 ... 2.104e7] Pa

$$P_sat_curve := \overrightarrow{\text{IF97_psat_T}(T_sat_nodes)}$$

Saturation pressure in bar: [0.012 ... 210.43] bar

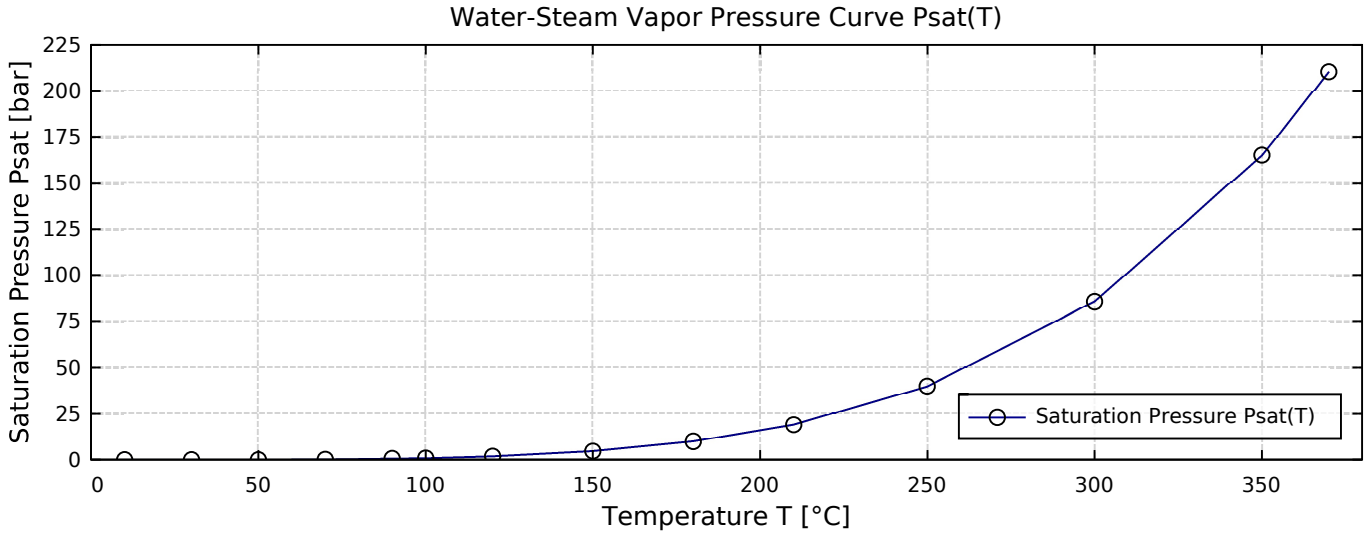
$$P_sat_bar := \frac{\overrightarrow{P_sat_curve}}{100000}$$

Saturation temperatures in Celsius: [10 ... 370] °C

$$T_sat_C := \overrightarrow{T_sat_nodes - 273.15\ \text{K}}$$

Vapor pressure curve coordinates (T [°C], Psat [bar]) for 2D XY plot

$$xy_psat_curve := \text{augment}(T_sat_C, P_sat_bar)$$



{xy_psat_curve

Not for commercial use

Created using a free version of SMath Studio

Section 2: Phase Diagram & Saturation Dome Visualization

Temperature-Entropy (T-s) diagram displaying the two-phase vapor-liquid equilibrium envelope.

Saturation pressure nodes: 11 points spanning 10 kPa to 22 MPa

$$P_dome := \begin{bmatrix} 10000 \\ 50000 \\ 100000 \\ 500000 \\ 1000000 \\ 2000000 \\ 5000000 \\ 10000000 \\ 15000000 \\ 20000000 \\ 22000000 \end{bmatrix}$$

Saturation temperatures: T_dome = [318.96 ... 646.86] K

$$T_dome := \overrightarrow{\mathrm{IF97_Tsat_p}(P_dome)}$$

Saturated liquid entropies: s_dome_f = [649.2 ... 4297.6] J/(kg·K)

$$s_dome_f := \overrightarrow{\mathrm{IF97_sliq_p}(P_dome)}$$

Saturated vapor entropies: s_dome_g = [8148.9 ... 4529.3] J/(kg·K)

$$s_dome_g := \overrightarrow{\mathrm{IF97_svap_p}(P_dome)}$$

Scaled liquid entropy coordinates: [0.649 ... 4.298] kJ/(kg·K)

$$s_f_kJ := \frac{\overrightarrow{s_dome_f}}{1000}$$

Scaled vapor entropy coordinates: [8.149 ... 4.529] kJ/(kg·K)

$$s_g_kJ := \frac{\overrightarrow{s_dome_g}}{1000}$$

Saturation temperatures in Celsius: [45.81 ... 373.71] °C

$$T_dome_C := \overrightarrow{T_dome - 273.15\ \mathrm{K}}$$

Saturated liquid coordinates (s_f, T) for 2D XY plot

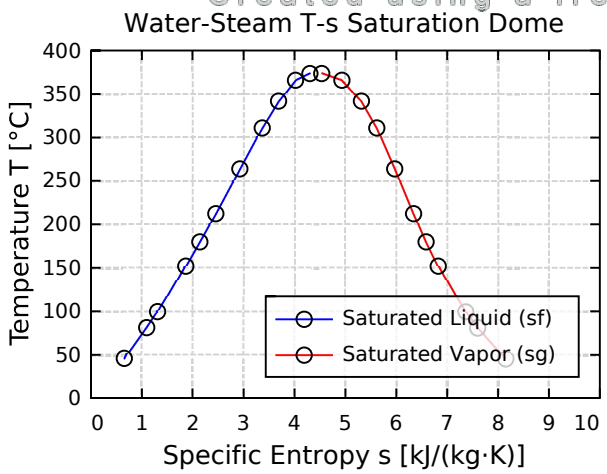
$$xy_dome_liq := \mathrm{augment}(s_f_kJ, T_dome_C)$$

Saturated vapor coordinates (s_g, T) for 2D XY plot

$$xy_dome_vap := \mathrm{augment}(s_g_kJ, T_dome_C)$$

Not for commercial use

Created using a free version of SMath Studio



$\begin{cases} xy_dome_liq \\ xy_dome_vap \end{cases}$

Section 3: Superheated Steam Forward Evaluation

State determination at turbine admission conditions: P = 4 MPa (40 bar), T = 400 °C (673.15 K).

Forward functions strictly take (T, p) as inputs.

Turbine admission pressure: P_sh = 4.0 MPa (40.0 bar)

$P_sh := 4000000\ Pa$

Turbine admission temperature: T_sh = 673.15 K (400.0 °C)

$T_sh := 400 + 273.15\ K$

Specific enthalpy: h_sh = 3214373.5 J/kg (3214.37 kJ/kg)

$h_sh := IF97_h_Tp(T_sh, P_sh) = 3.2144 \cdot 10^{6}\ Gy$

Specific entropy: s_sh = 6771.19 J/(kg·K) (6.7712 kJ/(kg·K))

$s_sh := IF97_s_Tp(T_sh, P_sh) = 6771.192\ \frac{Gy}{K}$

Specific internal energy: u_sh = 2920646.3 J/kg (2920.65 kJ/kg)

$u_sh := IF97_u_Tp(T_sh, P_sh) = 2.9206 \cdot 10^{6}\ Gy$

Steam density: rho_sh = 13.618 kg/m³

$rho_sh := IF97_rho_Tp(T_sh, P_sh) = 13.6181\ \frac{kg}{m^{3}}$

Specific volume: v_sh = 0.07343 m³/kg

$v_sh := IF97_v_Tp(T_sh, P_sh) = 0.0734\ \frac{m^{3}}{kg}$

Isobaric heat capacity: cp_sh = 2364.22 J/(kg·K) (2.364 kJ/(kg·K))

$cp_sh := IF97_cp_Tp(T_sh, P_sh) = 2364.2243\ \frac{Gy}{K}$

Not for commercial use

Created using a free version of SMath Studio

Isochoric heat capacity: cv_sh = 1733.70 J/(kg·K) (1.734 kJ/(kg·K))

$cv_sh := \text{IF97_cv_Tp}(T_sh, P_sh) = 1733.7029 \ \frac{Gy}{K}$

Speed of sound: w_sh = 614.49 m/s

$w_sh := \text{IF97_w_Tp}(T_sh, P_sh) = 614.4893 \ \frac{m}{s}$

Dynamic viscosity: mu_sh = 2.4403e-5 Pa·s (24.40 µPa·s)

$mu_sh := \text{IF97_mu_Tp}(T_sh, P_sh) = 2.4403 \cdot 10^{-5} \ s \ Pa$

Thermal conductivity: tc_sh = 0.05923 W/(m·K)

$tc_sh := \text{IF97_tc_Tp}(T_sh, P_sh) = 0.0592 \ \frac{N}{s \ K}$

Prandtl number: pr_sh = 0.9741

$pr_sh := \text{IF97_Pr_Tp}(T_sh, P_sh) = 0.9741$

Visualization 3: Superheated Steam Heat Capacity cp(T) at 40 bar (4 MPa)

Temperature dependence of cp from near saturation (260 °C) up to 600 °C, showing the sharp rise near saturation.

Temperature nodes for 40 bar isobar: [260 ... 600] °C (533.15 ... 873.15 K)

$$T_cp_nodes := \begin{bmatrix} 533.15 \\ 553.15 \\ 573.15 \\ 603.15 \\ 633.15 \\ 673.15 \\ 723.15 \\ 773.15 \\ 823.15 \\ 873.15 \end{bmatrix} K$$

Isobaric heat capacities at 4 MPa: [3553.6 ... 2298.2] J/(kg·K)

$cp_nodes := \overrightarrow{\text{IF97_cp_Tp}(T_cp_nodes, 4000000 \ Pa)}$

Scaled heat capacity in kJ/(kg·K): [3.554 ... 2.298] kJ/(kg·K)

$cp_nodes_kJ := \frac{\overrightarrow{cp_nodes}}{1000}$

Temperatures in Celsius: [260 ... 600] °C

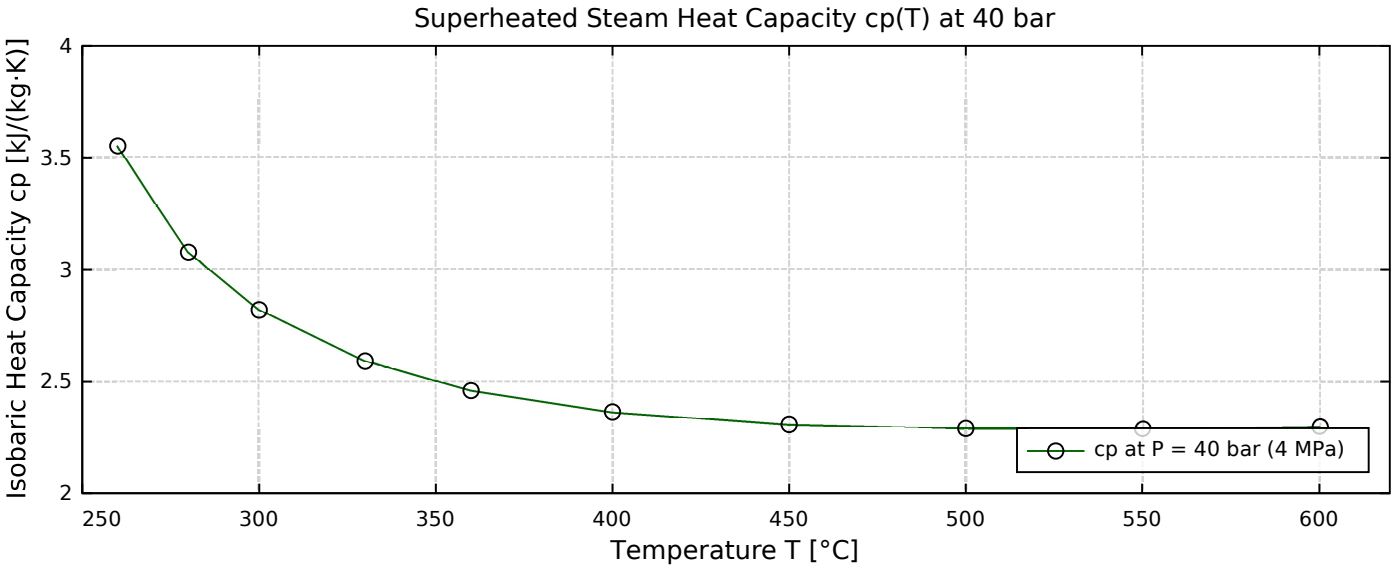
$T_cp_C := \overrightarrow{T_cp_nodes - 273.15 \ K}$

Isobaric heat capacity coordinates (T [°C], cp [kJ/(kg·K)]) at 40 bar

$xy_cp_isobar := \text{augment}(T_cp_C, cp_nodes_kJ)$

Not for commercial use

Created using a free version of SMath Studio



{xy_cp_isobar

Section 4: Backward Functions & State Inversion

Verification of IF97 dedicated backward equations T(p,h), rho(p,h), T(p,s), rho(p,s), p(h,s), T(h,s).

Backward temperature from (p, h): T_from_ph = 673.1434 K (recovers 673.15 K)

$T_from_ph := \mathrm{IF97_T_ph}(P_sh, h_sh) = 673.1434\ \mathrm{K}$

Backward density from (p, h): rho_from_ph = 13.618 kg/m³

$rho_from_ph := \mathrm{IF97_rho_ph}(P_sh, h_sh) = 13.6182\ \frac{\mathrm{kg}}{\mathrm{m}^3}$

Backward temperature from (p, s): T_from_ps = 673.1486 K (recovers 673.15 K)

$T_from_ps := \mathrm{IF97_T_ps}(P_sh, s_sh) = 673.1486\ \mathrm{K}$

Backward density from (p, s): rho_from_ps = 13.618 kg/m³

$rho_from_ps := \mathrm{IF97_rho_ps}(P_sh, s_sh) = 13.6181\ \frac{\mathrm{kg}}{\mathrm{m}^3}$

Backward pressure from (h, s): P_from_hs = 3999966.6 Pa (recovers 4.0 MPa)

$P_from_hs := \mathrm{IF97_p_hs}(h_sh, s_sh) = 4 \cdot 10^{6}\ \mathrm{Pa}$

Backward temperature from (h, s): T_from_hs = 673.1432 K (recovers 673.15 K)

$T_from_hs := \mathrm{IF97_T_hs}(h_sh, s_sh) = 673.1432\ \mathrm{K}$

IF97 Region from (p, h): reg_sh_ph = 2 (Superheated Vapor Region)

$reg_sh_ph := \mathrm{IF97_Region_ph}(P_sh, h_sh) = 2$

IF97 Region from (p, s): reg_sh_ps = 2 (Superheated Vapor Region)

$reg_sh_ps := \mathrm{IF97_Region_ps}(P_sh, s_sh) = 2$

Not for commercial use

Created using a free version of SMath Studio

Section 5: Wet Steam Mixture & Two-Phase Thermodynamics

Properties of two-phase liquid-vapor mixture at P = 2 MPa and specified vapor quality Q = 0.75

calculated directly via fundamental thermodynamic relations: M = M_liq + Q * (M_vap - M_liq).

Two-phase mixture pressure: P_mix = 2.0 MPa (20.0 bar)

$P_mix := 2000000\ \text{Pa}$

Specified dry steam vapor quality: Q_spec = 0.75 (75% vapor mass fraction)

$Q_spec := 0.75$

Saturated liquid enthalpy at 2 MPa: h_liq_mix = 908605.2 J/kg

$h_liq_mix := \text{IF97_hliq_p}(P_mix) = 9.0862 \cdot 10^{5}\ \text{Gy}$

Latent heat of vaporization at 2 MPa: h_fg_mix = 1889784.6 J/kg

$h_fg_mix := \text{IF97_hfg_p}(P_mix) = 1.8898 \cdot 10^{6}\ \text{Gy}$

Wet mixture enthalpy: h_mix = 2325943.6 J/kg (2325.94 kJ/kg)

$h_mix := h_liq_mix + Q_spec \cdot h_fg_mix = 2.3259 \cdot 10^{6}\ \text{Gy}$

Saturated liquid entropy at 2 MPa: s_liq_mix = 2447.34 J/(kg·K)

$s_liq_mix := \text{IF97_sliq_p}(P_mix) = 2447.0239\ \frac{\text{Gy}}{\text{K}}$

Saturated vapor entropy at 2 MPa: s_vap_mix = 6340.50 J/(kg·K)

$s_vap_mix := \text{IF97_svap_p}(P_mix) = 6339.1644\ \frac{\text{Gy}}{\text{K}}$

Wet mixture entropy: s_mix = 5367.21 J/(kg·K) (5.367 kJ/(kg·K))

$s_mix := s_liq_mix + Q_spec \cdot (s_vap_mix - s_liq_mix) = 5366.1293\ \frac{\text{Gy}}{\text{K}}$

Saturated liquid density at 2 MPa: rho_liq_mix = 852.79 kg/m³

$rho_liq_mix := \text{IF97_rholiq_p}(P_mix) = 849.798\ \frac{\text{kg}}{\text{m}^3}$

Saturated vapor density at 2 MPa: rho_vap_mix = 10.047 kg/m³

$rho_vap_mix := \text{IF97_rhovap_p}(P_mix) = 10.0421\ \frac{\text{kg}}{\text{m}^3}$

Wet mixture specific volume: v_mix = 0.07494 m³/kg

$v_mix := \frac{1 - Q_spec}{rho_liq_mix} + \frac{Q_spec}{rho_vap_mix} = 0.075\ \frac{\text{m}^3}{\text{kg}}$

Wet mixture density: rho_mix = 13.344 kg/m³

$rho_mix := \frac{1}{v_mix} = 13.337\ \frac{\text{kg}}{\text{m}^3}$

Recovered vapor quality from (p, h): Q_from_ph = 0.7500 (100% exact)

$Q_from_ph := \frac{h_mix - h_liq_mix}{h_fg_mix} = 0.75$

Not for commercial use

Created using a free version of SMath Studio

Recovered vapor quality from (p, s): Q_from_ps = 0.7500 (100% exact)

$$Q_from_ps := \frac{s_mix - s_liq_mix}{s_vap_mix - s_liq_mix} = 0.75$$

Section 6: Temperature-Based Saturation & Extended Transport Properties

Verification of temperature-driven saturation functions, speed of sound, and heat capacities.

Evaluation temperature: T_sat_test = 423.15 K (150.0 °C)

$$T_sat_test := 150 + 273.15\ \mathrm{K}$$

Saturated liquid enthalpy from T: h_liq_at_T = 632198.8 J/kg (632.20 kJ/kg)

$$h_liq_at_T := \mathrm{IF97_hliq_T}(T_sat_test)$$

Saturated vapor enthalpy from T: h_vap_at_T = 2745914.8 J/kg (2745.91 kJ/kg)

$$h_vap_at_T := \mathrm{IF97_hvap_T}(T_sat_test)$$

Latent heat of vaporization from T: h_fg_at_T = 2113716.0 J/kg (2113.72 kJ/kg)

$$h_fg_at_T := \mathrm{IF97_hfg_T}(T_sat_test)$$

Saturated liquid speed of sound at 1 MPa: w_liq_1MPa = 1386.41 m/s

$$w_liq_1MPa := \mathrm{IF97_wliq_p}(P_sat_1MPa) = 1391.6387\ \frac{\mathrm{m}}{\mathrm{s}}$$

Saturated vapor speed of sound at 1 MPa: w_vap_1MPa = 512.43 m/s

$$w_vap_1MPa := \mathrm{IF97_wvap_p}(P_sat_1MPa) = 500.8939\ \frac{\mathrm{m}}{\mathrm{s}}$$

Saturated liquid isobaric heat capacity: cp_liq_1MPa = 4423.82 J/(kg·K)

$$cp_liq_1MPa := \mathrm{IF97_cpliq_p}(P_sat_1MPa) = 4405.112\ \frac{\mathrm{Gy}}{\mathrm{K}}$$

Saturated vapor isobaric heat capacity: cp_vap_1MPa = 2697.55 J/(kg·K)

$$cp_vap_1MPa := \mathrm{IF97_cpvap_p}(P_sat_1MPa) = 2714.9848\ \frac{\mathrm{Gy}}{\mathrm{K}}$$

Saturated liquid Prandtl number: Pr_liq_1MPa = 0.9916

$$Pr_liq_1MPa := \mathrm{IF97_Prliq_p}(P_sat_1MPa) = 0.9874$$

Saturated vapor Prandtl number: Pr_vap_1MPa = 1.1624

$$Pr_vap_1MPa := \mathrm{IF97_Prvap_p}(P_sat_1MPa) = 1.1684$$

Section 7: Industrial Case Study — Rankine Cycle Steam Turbine Expansion

High-pressure turbine expansion from 8 MPa, 520 °C down to condenser vacuum (8 kPa) with eta_turb = 88%.

Includes Mollier (h-s) Diagram visualization with the steam expansion trajectory.

High-pressure turbine inlet: P_turb_in = 8.0 MPa (80.0 bar)

$$P_turb_in := 8000000\ \mathrm{Pa}$$

Not for commercial use

Created using a free version of SMath Studio

Turbine inlet steam temperature: T_turb_in = 793.15 K (520.0 °C)

$T_turb_in := 520 + 273.15\ \mathrm{K}$

Turbine inlet enthalpy: h_turb_in = 3448640.8 J/kg (3448.64 kJ/kg)

$h_turb_in := \mathrm{IF97_h_Tp}(T_turb_in, P_turb_in) = 3.4486 \cdot 10^{6}\ \mathrm{Gy}$

Turbine inlet entropy: s_turb_in = 6789.29 J/(kg·K) (6.7893 kJ/(kg·K))

$s_turb_in := \mathrm{IF97_s_Tp}(T_turb_in, P_turb_in) = 6789.2941\ \frac{\mathrm{Gy}}{\mathrm{K}}$

Condenser operating vacuum pressure: P_cond = 8000 Pa (0.080 bar, 8.0 kPa)

$P_cond := 8000\ \mathrm{Pa}$

Ideal isentropic expansion: s_turb_out_s = 6789.29 J/(kg·K)

$s_turb_out_s := s_turb_in$

Saturated liquid entropy at condenser pressure: s_cond_f = 559.08 J/(kg·K)

$s_cond_f := \mathrm{IF97_sliq_p}(P_cond) = 592.5316\ \frac{\mathrm{Gy}}{\mathrm{K}}$

Saturated vapor entropy at condenser pressure: s_cond_g = 8246.33 J/(kg·K)

$s_cond_g := \mathrm{IF97_svap_p}(P_cond) = 8227.4109\ \frac{\mathrm{Gy}}{\mathrm{K}}$

Saturated liquid enthalpy at condenser pressure: h_cond_f = 173846.5 J/kg

$h_cond_f := \mathrm{IF97_hliq_p}(P_cond) = 1.7385 \cdot 10^{5}\ \mathrm{Gy}$

Latent heat at condenser pressure: h_cond_fg = 2403144.2 J/kg

$h_cond_fg := \mathrm{IF97_hfg_p}(P_cond) = 2.4024 \cdot 10^{6}\ \mathrm{Gy}$

Isentropic exhaust vapor quality: Q_turb_out_s = 0.8105 (81.05% dry steam)

$Q_turb_out_s := \frac{s_turb_out_s - s_cond_f}{s_cond_g - s_cond_f} = 0.8116$

Isentropic exhaust enthalpy: h_turb_out_s = 2121575.4 J/kg (2121.58 kJ/kg)

$h_turb_out_s := h_cond_f + Q_turb_out_s \cdot h_cond_fg = 2.1237 \cdot 10^{6}\ \mathrm{Gy}$

Isentropic specific turbine work: w_turb_s = 1327065.4 J/kg (1327.07 kJ/kg)

$w_turb_s := h_turb_in - h_turb_out_s = 1.3249 \cdot 10^{6}\ \mathrm{Gy}$

Internal isentropic turbine efficiency: eta_turb = 0.88 (88.0%)

$eta_turb := 0.88$

Actual specific shaft work: w_turb_act = 1167817.5 J/kg (1167.82 kJ/kg)

$w_turb_act := w_turb_s \cdot eta_turb = 1.1659 \cdot 10^{6}\ \mathrm{Gy}$

Actual turbine exhaust enthalpy: h_turb_out = 2280823.3 J/kg (2280.82 kJ/kg)

$h_turb_out := h_turb_in - w_turb_act = 2.2827 \cdot 10^{6}\ \mathrm{Gy}$

Not for commercial use

Created using a free version of SMath Studio

Actual turbine exhaust vapor quality: Q_turb_out = 0.8768 (87.68% dry steam)

$$Q_turb_out := \frac{h_turb_out - h_cond_f}{h_cond_fg} = 0.8778$$

Turbine exhaust condensing temperature: T_turb_out = 314.6601 K (41.51 °C)

$$T_turb_out := \text{IF97_Tsat_p}(P_cond) = 314.6601\ \text{K}$$

Main steam mass flow rate: m_dot = 50.0 kg/s (180.0 t/h)

$$m_dot := \frac{50\ \text{kg}}{\text{s}}$$

Mechanical shaft power output: P_mech = 58390875.0 W (58.39 MW)

$$P_mech := m_dot \cdot w_turb_act = 5.8296 \cdot 10^{7}\ \text{W}$$

Electrical generator efficiency: eta_gen = 0.98 (98.0%)

$$eta_gen := 0.98$$

Net electrical generator power: P_elec = 57223057.5 W (57.22 MWe)

$$P_elec := P_mech \cdot eta_gen = 5.7131 \cdot 10^{7}\ \text{W}$$

Visualization 4: Mollier (h-s) Diagram & Turbine Expansion Trajectory

Enthalpy-Entropy chart showing saturated liquid (hf) and saturated vapor (hg) dome

alongside the irreversible turbine expansion path from admission (8 MPa, 520 °C) to condenser (8 kPa).

Saturated liquid enthalpies: h_dome_f = [1.918e5 ... 2.013e6] J/kg

$$h_dome_f := \overrightarrow{\text{IF97_hliq_p}(P_dome)}$$

Saturated vapor enthalpies: h_dome_g = [2.584e6 ... 2.163e6] J/kg

$$h_dome_g := \overrightarrow{\text{IF97_hvap_p}(P_dome)}$$

Scaled liquid enthalpy in kJ/kg: [191.8 ... 2013.4] kJ/kg

$$h_f_kJ := \frac{\overrightarrow{h_dome_f}}{1000}$$

Scaled vapor enthalpy in kJ/kg: [2583.9 ... 2163.2] kJ/kg

$$h_g_kJ := \frac{\overrightarrow{h_dome_g}}{1000}$$

Mollier saturated liquid curve (sf, hf)

$$xy_mollier_liq := \text{augment}(s_f_kJ, h_f_kJ)$$

Mollier saturated vapor curve (sg, hg)

$$xy_mollier_vap := \text{augment}(s_g_kJ, h_g_kJ)$$

Actual turbine exhaust entropy: s_turb_out_act = 7298.54 J/(kg·K)

$$s_turb_out_act := s_cond_f + Q_turb_out \cdot s_cond_g - s_cond_f$$

Not for commercial use

Created using a free version of SMath Studio

Turbine expansion entropy path: inlet 6.789 to exit 7.299 kJ/(kg·K)

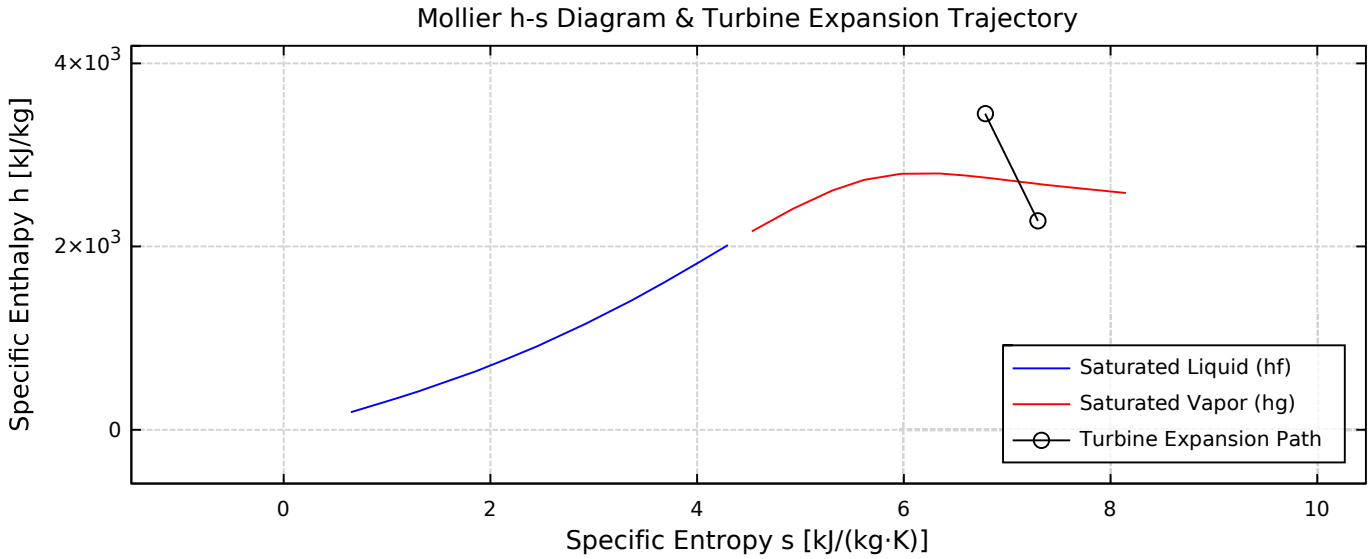
$$s_turb_path := \begin{bmatrix} 6.78929 \\ 7.29854 \end{bmatrix}$$

Turbine expansion enthalpy path: inlet 3448.6 to exit 2280.8 kJ/kg

$$h_turb_path := \begin{bmatrix} 3448.64 \\ 2280.82 \end{bmatrix}$$

Turbine expansion line (s, h) on Mollier diagram

$xy_turb_exp := \text{augment}(s_turb_path, h_turb_path)$



$$\begin{cases} xy_mollier_liq \\ xy_mollier_vap \\ xy_turb_exp \end{cases}$$

Not for commercial use